gemstone-py Cookbook

Task-focused recipes for daily use

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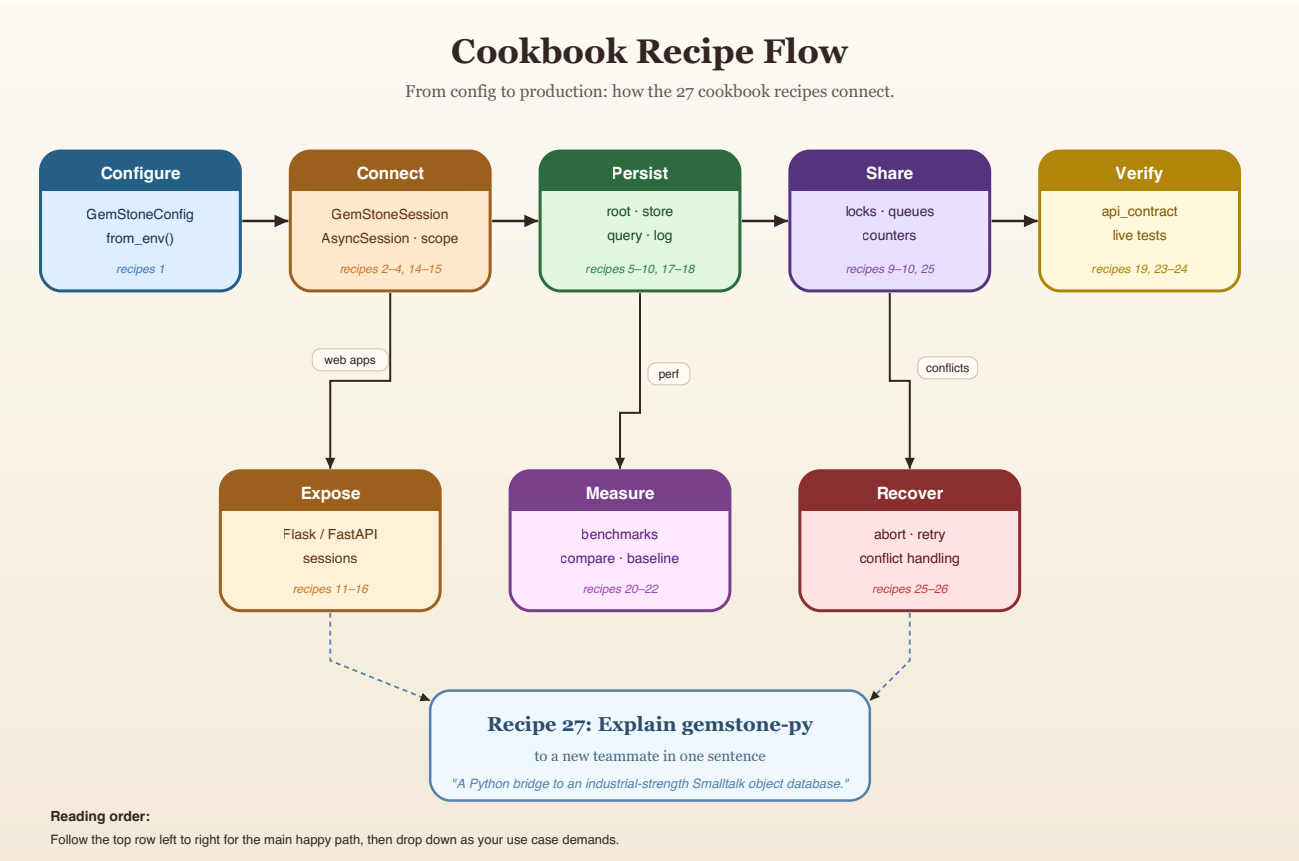
Contents

Cookbook

This cookbook is a collection of direct recipes. Each one is deliberately short enough to copy, adapt, and keep moving.



Recipe 1: Open a Session and Evaluate Smalltalk

```
from gemstone_py import GemStoneConfig, GemStoneSession

config = GemStoneConfig.from_env()

with GemStoneSession(config=config) as session:
    print(session.eval("3 factorial"))
```

Use this when:

- you want the smallest live sanity check
- you need to inspect a repository value quickly

Recipe 2: Commit a Write Safely

```
from gemstone_py import GemStoneConfig, TransactionPolicy, GemStoneSession
from gemstone_py.persistent_root import PersistentRoot

config = GemStoneConfig.from_env()

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    root = PersistentRoot(session)
    root["CookbookExample"] = {"answer": 42}
```

Why this recipe exists:

- it avoids the classic "the write seemed to work but vanished" mistake

Recipe 3: Read From `PersistentRoot`

```
from gemstone_py import GemStoneSession
from gemstone_py.persistent_root import PersistentRoot

with GemStoneSession(config=config) as session:
    root = PersistentRoot(session)
    print(root["CookbookExample"])
```

Recipe 4: Use `session_scope(...)`

```
from gemstone_py import GemStoneConfig, session_scope
from gemstone_py.persistent_root import PersistentRoot

config = GemStoneConfig.from_env()

with session_scope(config=config) as session:
    root = PersistentRoot(session)
    root["ScopedWork"] = {"kind": "unit-of-work"}
```

This is the recipe to prefer in application code.

Recipe 5: Create a Named `GSCollection`

```
from gemstone_py.gsquery import GSCollection

people = GSCollection("CookbookPeople", config=config)
people.insert({"@name": "Ada", "@city": "London"})
people.insert({"@name": "Grace", "@city": "New York"})
people.add_index("@city")
```

Recipe 6: Search an Indexed Collection

```
matches = people.search("@city", "eql", "London")
for row in matches:
    print(row)
```

Use `GSCollection` when you need repeated search, not when you merely enjoy the idea of repeated search.

Recipe 7: Keep a Store-Like Dataset in `GStore`

```
from gemstone_py.gstore import GStore

store = GStore("cookbook.db", config=config)
store["sku:hat"] = {"name": "Hat", "stock": 8}
store["sku:cape"] = {"name": "Cape", "stock": 3}
print(store["sku:hat"])
```

Recipe 8: Append a Persistent Event Log Entry

```
from gemstone_py.objectlog import ObjectLog

log = ObjectLog(config=config)
log.info("inventory_adjusted", {"sku": "hat", "delta": -1})
```

Later:

```
for entry in log.entries():
    print(entry)
```

Recipe 9: Share a Counter Between Sessions

```
from gemstone_py.concurrency import RCCounter

with session_scope(config=config) as session:
    counter = RCCounter(session)
    counter += 1
```

Use shared counters when the number truly lives in GemStone, not when a local integer would do and you are simply feeling theatrical.

Recipe 10: Share a Queue

```
from gemstone_py.concurrency import RCQueue
from gemstone_py.persistent_root import PersistentRoot

with session_scope(config=config) as session:
    root = PersistentRoot(session)
    root["WorkQueue"] = RCQueue(session)
```

Recipe 11: Install Flask Request Sessions

```
from flask import Flask
from gemstone_py import GemStoneConfig
from gemstone_py.frameworks.flask import install_flask_request_session

app = Flask(__name__)

install_flask_request_session(
    app,
    config=GemStoneConfig.from_env(),
    pool_size=4,
)
```

This is the default production-friendly shape.

Recipe 12: Use a Thread-Local Provider for Simpler Hosting

```
install_flask_request_session(
    app,
    config=GemStoneConfig.from_env(),
```

```
    thread_local=True,
)
```

Good when the server model is simple and you want one session per thread.

Recipe 13: Inspect Provider Health

```
from gemstone_py import flask_request_session_provider_snapshot

@app.get("/health/gemstone")
def gemstone_health():
    return flask_request_session_provider_snapshot()
```

Operational visibility is better than optimism.

Recipe 14: Build a Custom Web Adapter

```
from gemstone_py import GemStoneConfig, RequestScope, TransactionPolicy
from gemstone_py.session_providers import GemStoneSessionPool

pool = GemStoneSessionPool(maxsize=4, config=GemStoneConfig.from_env())

scope = RequestScope(
    session_provider=pool,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
)
session = scope.session()
try:
    session.eval("3 + 4")
finally:
    scope.finalize()
```

`RequestScope` is the framework-neutral sync lifecycle primitive behind Flask request sessions. For async frameworks, pair `AsyncRequestScope` with `AsyncSessionPool` and await `scope.finalize(...)` from request cleanup.

Recipe 15: Open an Async Session

```
from gemstone_py import GemStoneConfig
from gemstone_py.aio import AsyncSession

config = GemStoneConfig.from_env()
```

```
async with AsyncSession.connect(config=config) as session:
    print(await session.eval("3 + 4"))
```

Use this when the surrounding application is already async.

Recipe 16: Use an Async Transaction

```
async with AsyncSession.connect(config=config) as session:
    async with session.transaction():
        root = session.root()
        await root.set("AsyncCookbook", {"status": "ok"})
```

The transaction context commits on success and aborts on exception.

Recipe 17: Add FastAPI Dependency Injection

```
from fastapi import Depends, FastAPI
from gemstone_py import GemStoneConfig
from gemstone_py.aio import AsyncSession
from gemstone_py.aio.fastapi import session_dependency

app = FastAPI()
get_gemstone = session_dependency(config=GemStoneConfig.from_env())

@app.get("/health/gemstone")
async def gemstone_health(session: AsyncSession = Depends(get_gemstone)):
    return {"result": await session.eval("3 + 4")}
```

Django apps use the sync middleware adapter:

```
from django.http import JsonResponse
from gemstone_py import GemStoneConfig
from gemstone_py.frameworks.django import GemStoneSessionMiddleware, request_session

def gemstone_session_middleware(get_response):
    return GemStoneSessionMiddleware(get_response, config=GemStoneConfig.from_env())

def gemstone_health(request):
    session = request_session(request)
    return JsonResponse({"result": session.eval("3 + 4")})
```

For Litestar, use the sibling async adapter:

```
from litestar import Litestar, get
from litestar.di import Provide
```

```
from gemstone_py import GemStoneConfig
from gemstone_py.aio.litestar import session_dependency

get_gemstone = session_dependency(config=GemStoneConfig.from_env())

@get("/health/gemstone", dependencies={"session": Provide(get_gemstone)})
async def gemstone_health(session):
    return {"result": await session.eval("3 + 4")}

app = Litestar(route_handlers=[gemstone_health])
```

Recipe 18: Query With a Typed Protocol

```
from typing import Protocol
from gemstone_py.gsquery import GSCollection

class PostRecord(Protocol):
    title: str
    status: str

posts = GSCollection("SimplePosts", config=config).query(PostRecord)
for post in posts.where(lambda row: row.status == "published").all():
    print(post.title)
```

The lambda records an indexed GemStone path. It is not called for every row in Python.

Recipe 19: Generate a Typed Smalltalk Wrapper

```
from typing import Protocol
from gemstone_py import gemstone_class, gemstone_selector

@gemstone_class("OkzBooking", async_=True)
class OkzBookingProto(Protocol):
    status: str

    @classmethod
    @gemstone_selector("findById:")
    def find_by_id(cls, booking_id: str) -> "OkzBookingProto":
        ...

    def mark_paid(self, at_posix_seconds: int) -> None:
        ...
```

Generate and check in the wrapper:

```
gemstone-codegen \
  --module examples.typed_access.codegen_demo.models \
  --output examples/typed_access/codegen_demo/generated \
  --clean
```

Use it without hand-writing the class-side Smalltalk string:

```
from examples.typed_access.codegen_demo.generated import OkzBooking

booking = OkzBooking.find_by_id(session, "B-1001")
print(booking.status)
booking.mark_paid(1_779_912_000)
```

Recipe 20: Keep an OOP Alive Explicitly

```
with GemStoneSession(config=config) as session:
    ref = session.execute_managed("OrderedCollection new")
    ref.send("add:", session.new_string("kept"))
    print(ref.print_string())
    ref.close()
```

For a raw OOP:

```
with session.handle(raw_oop) as handle:
    print(handle.send("printString"))
```

Recipe 21: Check the Native Backend

```
python -m pip install "gemstone-py[fast]"
python -m examples.native_backend.check_backend
```

Force a backend before Python starts:

```
GEMSTONE_PY_GCI_BACKEND=ctypes python -m examples.native_backend.check_backend
GEMSTONE_PY_GCI_BACKEND=native python -m examples.native_backend.check_backend
```

Recipe 22: Run the Maintained Benchmarks

```
gemstone-benchmarks --entries 500 --search-runs 20
```

Or emit JSON:

```
gemstone-benchmarks --json --output benchmark-report.json
```

Recipe 23: Compare Benchmark Reports

```
gemstone-benchmark-compare old.json new.json --json --output compare.json
```

That turns performance arguments into evidence, which is disappointingly healthy.

Recipe 24: Register a New Accepted Benchmark Baseline

```
gemstone-benchmark-baseline-register \
  benchmark-report.json \
  --manifest .github/benchmarks/index.json
```

Recipe 25: Verify the Installed Artifact

```
python -m gemstone_py.api_contract --json
```

This is useful after installation, after release, and after any moment when you feel your package metadata may have become sentient.

For a full PyPI/TestPyPI release check:

```
gemstone-publish-verify --gemstone-version 0.2.12 --native-version 0.1.2
```

That command checks project JSON, version-specific JSON, the simple index, and temporary-virtualenv installs for both the pure package and native package.

Recipe 26: Scaffold and Run Module-Style Migrations

Create a numbered migration file:

```
gemstone-migrations scaffold add_amount_to_booking --directory migrations
```

A migration module exposes `upgrade(session)`, optional `downgrade(session)`, and optional `dependencies`:

```
id = "002_add_amount_to_booking"
dependencies = ("001_initial",)

def upgrade(session):
    session.eval(
        "OkzBooking addInstVarName: 'amount' ifAbsent: [ nil ]"
    )
    session.commit()

def downgrade(session):
    session.eval(
        "OkzBooking removeInstVarName: 'amount' ifAbsent: [ nil ]"
    )
    session.commit()
```

Register modules in a manifest and apply them:

```
from gemstone_py import GemStoneConfig, GemStoneSession
from gemstone_py.migrations import current_version, load_manifest, upgrade

steps = load_manifest("my_app.migrations.manifest")

with GemStoneSession(config=GemStoneConfig.from_env()) as session:
    print(current_version(session))
    print(upgrade(session, steps, dry_run=True))
    upgrade(session, steps)
```

Applied versions are recorded under `GemstonePyMigrations` in `UserGlobals`. The runner refuses to continue when the stone has applied migrations that are missing from the local manifest, or when a stored migration checksum differs from the local file. Real `upgrade` and `downgrade` runs acquire an advisory lock in `UserGlobals` before applying steps; dry-runs do not.

The same operations are available from the command line:

```
gemstone-migrations current
gemstone-migrations status --manifest my_app.migrations.manifest
gemstone-migrations plan --manifest my_app.migrations.manifest
```

```
gemstone-migrations upgrade --manifest my_app.migrations.manifest --dry-run
gemstone-migrations upgrade --manifest my_app.migrations.manifest --dry-run --record
gemstone-migrations upgrade --manifest my_app.migrations.manifest
gemstone-migrations downgrade --manifest my_app.migrations.manifest --target
001_initial
```

For a stale lock left by a crashed process, use a deliberate override:

```
gemstone-migrations upgrade --manifest my_app.migrations.manifest \
  --force-lock --lock-owner release-2026-05-11
```

`--dry-run --record` uses a recording session for callbacks and prints common session calls such as `session.eval(...)`, `execute(...)`, `perform_value(...)`, `commit()`, and `abort()` without sending them to GemStone.

Example recorded output:

```
dry-run upgrade: 1 step(s)
  002_add_amount_to_booking
recorded operations:
  # upgrade 002_add_amount_to_booking
  session.eval("OkzBooking addInstVarName: 'amount' ifAbsent: [ nil ]")
  session.commit()
```

Treat recorded dry-runs as a release review aid. Migrations that depend on live query results still need a real staging run because the recorder returns placeholder OOPs and never reads from GemStone.

To compare a local Protocol or type witness with the live GemStone class before writing the next migration:

```
gemstone-migrations diff-class OkzBooking \
  --local-class my_app.models:OkzBookingProto
```

The output lists missing and extra instance variables and prints advisory `session.eval(...)` lines for a reviewed migration.

Recipe 27: Run the Live Test Lane

```
GS_RUN_LIVE=1 ./scripts/run_live_checks.sh
```

Longer soak run:

```
GS_RUN_LIVE=1 GS_RUN_LIVE_SOAK=1 ./scripts/run_live_checks.sh
```

Recipe 28: Handle Commit Conflicts Without Pretending They Are Rare

When multiple sessions modify overlapping state, conflicts are normal. The right pattern:

```
from gemstone_py import GemStoneConfig, PersistentRoot, retrying_transaction

config = GemStoneConfig.from_env()

def increment_counter(session):
    # reload data each attempt; previous reads are stale after abort
    root = PersistentRoot(session)
    root["counter"] = (root.get("counter") or 0) + 1
    return root["counter"]

value = retrying_transaction(increment_counter, config=config, attempts=5)
```

Rules:

- keep the write unit small
- reload data after every abort — the previous read is stale
- bound the retry count — do not loop forever
- log conflicts — frequent conflicts signal a design smell, not bad luck

Add `on_conflict=` when you want a readable report for each retry:

```
def log_retry(retry):
    print(retry.format(include_summaries=False))

retrying_transaction(
    increment_counter,
    config=config,
    attempts=5,
    on_conflict=log_retry,
)
```

Recipe 29: Learn a Queue With a Hat

The hat trick example is memorable because it teaches a real primitive through a slightly ridiculous scenario. You should keep more examples like that in your own codebase than you probably do.

Recipe 30: Explain `gemstone-py` to a New Teammate

Use this sentence:

> "It is a Python package that talks directly to GemStone Smalltalk, keeps transactions explicit, gives us persistence helpers instead of a fake ORM, supports async, typed access, managed OOP lifetimes, and optional native wheels, and already has real CI, release, benchmark, and live verification lanes."

That sentence has rescued several meetings already. At minimum, it should rescue yours.

Recipe 31: Convert Scalar Values Explicitly

```
from datetime import date
from decimal import Decimal
from gemstone_py import GemStoneConfig, GemStoneSession,
scalar_value_converter_registry

registry = scalar_value_converter_registry()

with GemStoneSession(config=GemStoneConfig.from_env()) as session:
    invoice_oop = session.eval_oop("UserGlobals at: #CurrentInvoice")
    due_on, amount = registry.to_oops(session, [date(2026, 5, 15), Decimal("19.95")])
    session.perform_oop(invoice_oop, "dueOn:amount:", due_on, amount)
```

The built-in factories cover `datetime`, exact `date`, `Decimal`, and `UUID`. `ValueConverterRegistry.copy()` and `extend(...)` make application-specific registries cheap to assemble without global state.

For dataclasses, convert to a plain payload when that is what the GemStone-side API expects:

```
from gemstone_py import dataclass_to_dict

payload = dataclass_to_dict(booking_patch, recurse=False)
root["PendingBookingPatch"] = payload
```

Run the offline preview without a live stone:

```
gemstone-examples value-converters
python -m examples.cookbook.value_converters
```

Recipe 32: Batch Root and Dictionary Access

```
from gemstone_py import GemStoneConfig, PersistentRoot, session_scope

config = GemStoneConfig.from_env()

with session_scope(config=config) as session:
    root = PersistentRoot(session)

    values = root.get_many(["Settings", "VisitCount"], default=None)
    root.update_many({
        "VisitCount": int(values["VisitCount"] or 0) + 1,
        "LastMaintenance": "2026-05-14T22:00:00Z",
    })

    if "Settings" not in root:
        root["Settings"] = {}
    settings = root["Settings"]
    settings.update_many({"theme": "dark", "page_size": 50})
```

Use this when the operation is naturally a batch. It is still explicit repository access, just less chatty.

For persistent ordered lists:

```
from gemstone_py import PersistentRoot, session_scope
from gemstone_py.ordered_collection import OrderedCollection

with session_scope(config=config) as session:
    states = OrderedCollection(session)
    states.extend(["draft", "review", "paid"])
    PersistentRoot(session)["InvoiceStates"] = states
```

Recipe 33: Send Selectors in Bulk

```
from gemstone_py import GemStoneConfig, GemStoneSession, PerformCall

with GemStoneSession(config=GemStoneConfig.from_env()) as session:
    orders = [
        session.global_get("OrderA"),
        session.global_get("OrderB"),
    ]

    statuses = session.perform_many_value(orders, "status")
    labels = session.bulk_perform_calls_value([
        PerformCall(orders[0], "printString"),
        (orders[1], "status"),
    ])
```

Use `perform_many_` *for one selector across many receivers. Use* `bulk_perform_calls_` when the receivers, selectors, or arguments differ.

Recipe 34: Fingerprint Expected Schema

```
gemstone-migrations fingerprint \
  --root Bookings \
  --root BookingIndexes \
  --class OkzBooking \
  --manifest my_app.migrations.manifest \
  --json
```

For startup checks:

```
from gemstone_py.migrations import assert_schema_fingerprint, load_manifest

steps = load_manifest("my_app.migrations.manifest")
assert_schema_fingerprint(
    session,
    expected_digest="...",
    root_keys=["Bookings", "BookingIndexes"],
    class_names=["OkzBooking"],
    migration_steps=steps,
)
```

Fingerprinting is for deployment confidence. It does not apply migrations or infer Python object mappings.

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